Advanced Calculus I Non-hand-in Assignment 5

This homework comes from textbook and will **NOT** be collected (這份作業完成後不需要繳交、評分。). But for practice,or quizzes or exams, please make sure that you have idea to solve the following questions by yourself. (這些題目是很重要的練習，小考、期中、期末考會有類似的題目。練習這些題目的過程中所逐漸形成我們對於所學到的知識的"看法"，會在考試中很有幫助。所以，請一定要了解如何解題與解題背後的想法是什麼。)

Please prove the following propositions are true. (請證明下列命題爲眞。)

Section 2.2.
2.9 Theorem (ii) (on page 60)
2.11 Theorem (on page 61)

Section 9.5.
Suppose that $\{x_k\}$ and $\{y_k\}$ are sequences of points in $\mathbb{R}^n$. Also, $\{x_k\}$ and $\{y_k\}$ converge. Please show that

$$\lim_{k\to\infty}(x_k + y_k) = \lim_{k\to\infty} x_k + \lim_{k\to\infty} y_k$$